# Final Year Project

# AI Powered Analytics Web App for Smarter Investments

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# Chapter 1

## EXECUTIVE SUMMARY

Pakistan is in a transformative phase in financial technology, where the unprecedented pace of digitalization, changing investor behavior, and the increasing need to make decisions in real-time and driven by information make monitoring financial technologies increasingly important. These developments notwithstanding, a major information asymmetry still supports the lack of growth and involvement of retail investors in the Pakistan Stock Exchange (PSX). Although financial data, news outlets and analytical software are available in fragmented formats, the common investor still finds it hard to interpret, contextualize or synthesize such data to make an effective investment. This lack of connection brings up a key problem: more financial information is available, but there has been no corresponding increase in the capacity to use it and interpret it.

FinForecaster.pk Pakistan is the proposed solution that will fill this gap: it is an AI-based financial intelligence platform. The site combines real-time PSX information, machine-learning prediction systems, sentiment analysis of major Pakistani news outlets, and powerful control consoles that create intelligible information out of fractured data. FinForecaster.pk Pakistan will provide modern analytics and accessible visualization to allow retail investors, university students, and novice-level traders to have the tools commonly used by financial experts.

The present Executive Summary provides the essence of the vision, mission, value proposition, business opportunity, strategic contention, financial viability and anticipated impact of FinForecaster.pk Pakistan. It forms the general summary of the entire business plan, which gives visibility on the business idea and the reasons why it was created.

## Introduction to the Business Idea.

The idea of FinForecaster.pk Pakistan is that of an all-embracing digital ecosystem that can help one to make financial decisions much easier in the realm of artificial intelligence. The platform works with a broad scope of the problems that are being experienced by market players:

**Information Overload**

The raw data are bombarding the investors who do not have the capability to analyze or interpret the trends.

1. **Fragmented Data Sources**

The data of PSX, news sentiment, and company financials are scattered on various platforms which makes them to be confused.

2. **Low Financial Literacy**

The earnings reports, the financial ratios, the risk assessment and the technical indicators are not known to many retail investors.

**3. Lack of Predictive Guidance**

The majority of existing platforms offer historical data but do not offer AI-driven predictions and sentiment-based predictions.

FinForecaster.pk Pakistan addresses these problems by providing a single solution to combine multiple streams of data into a single convenient platform that is supplemented with predictive analytics and natural language processing.

## Mission and Vision

### Mission Statement

To empower Pakistani investors, students and other financial enthusiasts by availing clear, actionable and AI based market intelligence that makes financial decision-making easier and encourages responsible investment practices.

### Vision Statement

To be the most reliable and the most popular AI-driven stock exchange intelligent solution in Pakistan, making financial literacy available, decipherable, and practical to every corner of the society.

## The Issue: Why Pakistan Should Have This Platform.

The retail investing sector in Pakistan is plagued by the combination of problems that both limit entry and expansion:

**1. Lack of Financial Data knowledge.**

The majority of the individuals are incapable of reading complicated financial publications or analyzing the market indications. The notions of earnings per share, liquidity ratios, sentiment, or risk exposure are not available to regular investors.

**2. High Relying on Informal Channels.**

WhatsApp rumors, social media groups, and unverified tips affect investor decisions, and in most instances, this leads to either losses, panic selling or speculative trading.

**3. Inaccessible, Easy-to-use Tools.**

Current PSX and brokerage solutions offer a small amount of analytics, demand higher aptitude to interpret, and lack sentiment-based insights.

**4. Poor Exposure to Applied Finance to Students.**

The academic institutions of Pakistan are taught theoretical finance but lack mechanisms to train students in real time market analysis and this decreases employability to data driven jobs.

**5. Lack of Predictive Models to Local Markets.**

Pakistani market patterns are not global, and PSX volatility, industry-specific responses and news-based sentiment need to be modeled locally.

**FinForecaster.pk** Pakistan is a direct solution to these problems through AI, which makes the behavior of the market simple and easy to understand and act on.

## The Solution: Major Features and Value Proposal.

FinForecaster.pk Pakistan is a product that aims to provide an entire spectrum of tools that would be accessible, accurate, and with real-time functionality. It has a value proposition based on five pillars:

**1. Artificial Intelligence Sentiment Analysis.**

The platform applies natural language processing (NLP) to screen news on sources such as Business Recorder, Dawn Business, and Geo Economy, and the updates are classified as positive, negative, or neutral. The investors get instant information on the way in which the current events can affect the stock performance.

**2. Forecasting Models with Machine Learning.**

Predictive algorithms examine past stock prices, volume/trades, industry performance as well as sentiment ratings to create buy/hold/sell signals. Such models are not binding trading recommendations but they give predictions that are probability-based, thus creating a sense of transparency and responsible use.

**3. Live PSX Data Incorporation.**

Live updates on:

- Stock prices
- Index movements (KSE-100, KSE-30, KMI-30)
- Sector performance
- Trading volumes
- Candlestick patterns

This real time information allows quick and informed investment.

**4. Financial statement aggregation.**

Company profiles include:

- Income statements

- Balance sheets
- Cash flow reports
- Key ratios (P/E, ROE, ROA, D/E)

The ones that were shown were accompanied by visual representations to ease interpretation.

Appendix A

**5. Educational Knowledge and Resources.**

Accessible to investors and students are:

- Tutorials
- Interactive glossary
- Scenario-based simulations
- Trend explanation modules
- Easy to understand financial terminology.

This is a learning layer that transforms the platform into a tool, as well as educational ecosystem.

## Business Opportunity and Market Potential.

The Pakistani investment market is not fully utilized. There are over 240 million people and only a small part of them are involved with formal equity markets. The most important signs are the signs of enormous potential:

- Increasing youth population with financial interests.
- Growth in internet trading accounts.
- Increased smartphone and internet penetration.
- Digital initiatives of the government.
- Growing freelance and entrepreneurial society.

**1. FinForecaster.pk is aimed at:**

- Retail investors who are moving towards formal markets.
- Finance and business students in universities.
- New investors trying to be guided systematically.
- Individuals who investigate the secondary sources of income.
- AI-oriented tools are more appealing to technologically inclined users.

This aspect of accessibility coupled with the analytics is a strong business offer that can access millions.

## Business Model and Revenue Strategy.

The business employs a freemium-to-premium business model, which is a free basic functionality with monetized advanced functionality available as paid subscriptions.

1. **Free Features:**
   - Basic dashboards
   - Limited sentiment insights
   - Real-time PSX charts
   - Educational content
   - Watchlist feature

2. **Premium Features**:
   - Infinite sentiment analysis
   - Advanced ML forecasts
   - Risk assessment reports
   - Downloadable analytics
   - Deep-dive financial metrics
   - Sectoral heatmaps
   - Optimization tools of portfolio

3. **Other avenues of revenue are:**

   - Models of university licensing
   - Corporate analytical dashboards
   - Investor training and paid workshops

It has a wide base of revenues that guarantees financial sustainability in the long run.

## Financial Background and Solvency.

The financial model of the platform is chosen to make the overhead as low as possible and the impact as maximum:

1. **Minimal Development and operation costs:**

The primary costs include:

- Software development
- Training and hosting of AI model.
- Product marketing and user acquisition.
- Legal compliance
- Maintenance and updates

Since the platform is digital, the expansion to thousands of users will not be proportional to costs.

**2. High-Profit Potential**

Digital services usually contain:

- Low marginal cost per user
- High scalability
- Constant recurring revenue by subscription.

Low fixed cost and increasing rates paid subscribers will give the company a break-even early.

## Impact and Social Value

The company FinForecaster.pk Pakistan is not only a business entity, but it has something to offer to the Pakistani economy and education.

**1. Brings about Financial Literacy:**

The site helps beginners to understand the intricate financial principles because it makes them easier to grasp.

**2. Minimizes Fraud and Speculation:**

Through organized information, users do not depend much on rumors or unconfirmed information.

**3. Enhances Academic Learning**

Real world market analytics are provided to students in a real world setting, which is the difference between theory and practice.

4. **Promotes Data-Based Decision-Making**.

The insights generated by AI promote analytical thinking, which encourages discipline in investments.

**5. Sustainability of National Digitalization Objectives.**

The platform is in line with the vision of the Digital Pakistan to provide high-end fintech solutions that are accessible and tailored to the local needs.

**Conclusion**

FinForecaster.pk Pakistan is a progressive project that aims to provide Pakistani investors and students with the instruments needed to survive in the contemporary financial markets. It converts unstructured, messy data into intelligible data that can be used by consumers by utilizing artificial intelligence. The platform has a great future of growth and influence with high market demand, the availability of prices, scalability, and a very clear competitive edge.

This business plan also expounds on the market dynamics, competitor analysis, financial estimates, operational plans and risks of this venture. Combined these elements prove that FinForecaster.pk Pakistan can not only be a viable enterprise, but also be in a strategic position to be a top financial intelligence site in Pakistan.

# CHAPTER 2

## INTRODUCTION / BUSINESS DESCRIPTION

### Introduction

The Pakistani financial services industry is also undergoing a substantial change due to the digital innovation, changing consumer needs, and improved access to online investment platforms. However, despite these developments there is a sharp discrepancy between the supply of financial data and the capacity of the ordinary investor to process it and make significant action on the information. The Pakistan Stock Exchange (PSX) is one of the oldest exchanges in South Asia that is not fully utilized by the locals. The lack of wide participation is caused by limited financial literacy, disjointed sources of data, and the absence of analysis tools that would be friendly to beginners.

The proposed solution to this structural problem is the FinForecaster.pk Pakistan, which is a technology-based financial intelligence hub whose core idea is to democratize data interpretation and empower investors, especially retail investors, university students, and the new entrants, with the actionable information provided by AI and machine learning technologies. This chapter offers a background knowledge about the business, what problem it addresses, why it was developed, and the main goals on which its operation philosophy is based.

### Identifying the Problem

An effective business is created as a result of a well-established issue that influences a large market. The issue in Pakistan does not consist in the fact that the amount of financial information PSX, regulatory organizations, brokerage companies, and media houses generate more information on the issue than ever. The actual issue is with the interpretation of such information.

**Major challenges that affect Pakistani Investors**.

**1. Financial Literacy Level**

Pakistan has a low financial literacy level of less than 25% despite the rising financial awareness campaigns in the country. A majority of the new investors are unable to interpret annual reports, analyses price changes and comprehend macroeconomic indicators. EBITDA, ROE, MACD or stochastic oscillators are words used in a technical way which seem to be difficult to understand.

2. **Disjointed Sources of information**

An average investor needs to look through various sites in a day PSX portal to check prices, Business Recorder to view news, brokerage apps to check chart, independent blogs to analyze, and social media to check rumors. Such fragmentation introduces a sense of confusion and slowness of decision-making.

3. **Lack of Forecasting Intelligence**

The existing market tools are capable of providing historical charts but fail to give data-driven forecasts and sentiment-driven indicators. Most investors tend to make decisions on guesses other than informed projections.

4. **Excessive reliance in the Informal Channels**

In Pakistan, hearsay, WhatsApp chats, and social media groups have an inappropriate influence on investor behavior. These unofficial mediums are usually used to disseminate false information culminating in emotional trading and loss of funds.

5. **Absence of Applied Learning Tools to students.**

Theoretical finance is taught in universities, and the students seldom access real-time tools to reflect the current market analysis tools. This knowledge gap makes the graduates of finance less employable and it limits practical learning.

6. **Poor access to Analytical Tools.**

Advanced instruments in the foreign markets Bloomberg Terminal, Trading View premium, Reuters Eikon are either too costly or fail to provide localized data on the Pakistani market.

## The Gap

Investors in Pakistan need:

- Developed and confident data.
- Simple, digestible insights
- Predictive analytics
- Sentiment analysis in real-time.
- Beginner-friendly tools
- A platform that is meant to match local financial behavior.

FinForecaster.pk Pakistan is the solution to this essential gap.

## Business Objectives

The success of FinForecaster.pk Pakistan is pegged on clear short-term and long-term goals.

1. **Short-Term Objectives (Year 1-2)**

- Create and release a complete AI-based platform based on data integration, dashboards, sentiment analysis, and forecasting capabilities.
- Grow to 30,000+ users and 10,000 paid subscribers with premiums.
- Collaborate with Pakistani universities to integrate the platform into the curricula of finance.
- Create a high online presence through awareness campaigns and influencing.
- Alter and optimize machine-learning models with platform-created data.

2. **Long-Term Objectives (Year 5-7)**
   - Access 1 million plus users of the platform with a diversified range of customers such as students, investors and corporate clients.
   - Implement innovative AI facilities, including sectoral predictions, risk heat maps and economic predictive modeling.
   - Extend the platform to regional markets such as GCC, South Asian stock markets.
   - Make the platform a prominent InfoTech-fintech hybrid, and take part in national financial literacy.
   - Create a white-label product to the brokerage firms and financial institutions.

## Business Description

FinForecaster.pk Pakistan is not simply an application, but a financial intelligence ecosystem. It takes the data gathered by PSX, company reports, and real-time sentiment of the news and runs it through machine learning algorithms that generate forecasts and insights. The platform is a mix of technology, finance, and education to provide a product that is specific to the financial market in Pakistan.

**Fundamental Elements of the Business.**

1. **Data Integration Layer**

In charge of sourcing, cleaning, and transforming:

- Real-time PSX market data
- Past price volume data.
- Financial statements
- Corporate announcements
- Business news articles

ETL pipelines are powered by Python and they are accurate and consistent.

2. **Machine Learning Layer**

The AI module focuses on:

- Trend prediction

- Volatility estimation
- Sentiment classification
- Multivariate regression equations.
- Time-series prediction (ARIMA, LSTM, Prophet)
- The models are trained on localized datasets of Pakistanis to capture the actual PSX behavior.

**3. Dashboard Layer and Visualization.**

The following makes user experience easier:

- Interactive charts
- Heatmaps
- Sentiment meters
- Forecast models
- Risk indicators

Dashboards are mobile-first friendly, given the habits of Pakistan towards the use of digital.

**4. Education Layer**

Includes:

- Tutorials
- Video lessons
- Glossaries
- Example-based explanations

This layer will make the platform a learning experience to beginners and students.

## Unique Selling Proposition (USP)

The localization, AI-driven insights, and user-centricity are the three things that FinForecaster.pk Pakistan distinguishes itself based on.

**Key USPs:**

**1. Specific AI predictions in Pakistan.**

In contrast to global tools, the algorithms of the platform are custom-trained on data on the Pakistani market, and it is based on this, which makes predictions closer to the contexts.

**2. Real-Time sentimental analysis on local news.**

The NLP engine extracts sentiment in Urdu and English financial news, and offers sentiment which is distinctly differentiated.

**3. Beginner-Friendly Visualizations**

The complicated financial indicators are represented visually in a form of color-coded, simplified and understandable.

**4. Educational Integration**

None of the PSX oriented platforms combines instructional content and real-time data.

## The way FinForecaster.pk can respond to the needs of Pakistan.

**1. Diligently Brings together Dispersed Financial Data**

All the essential information is available to investors in a single location: prices, news, sentiment, forecasts, and financial statements.

**2. Lessens a Dependence on Rumors and Speculation.**

The platform also prevents the use of emotions and rumor-based decisions, since it delivers factual information supported by AI.

**3. Makes PSX Approachable to Novices.**

The real-time examples can teach students and new investors the basics of the market rather than the old textbooks can teach them the old concepts.

**4. Offers Predictive Insights Inaccessible on any Current Platform.**

The models of the forecasts give the probabilities and ranges of the market, which give a realistic view of the market pattern.

**5. Enhances Decision-Making**

Investors are able to evaluate risk, monitor performance, sentiment, and forecast on a single dashboard.

## Pakistani Financial Ecosystem Long-term vision.

FinForecaster.pk Pakistan is in line with the more general purpose of national development:

**1. Digital Pakistan Initiative**

The platform promotes digital engagement in financial markets, which helps with the government-led efforts to transform digital.

**2. Economic Empowerment**

The platform promotes wealth creation and risk-taking with good knowledge by increasing financial literacy and decision-making.

**3. Academic Strengthening**

Colleges will be able to gain access to an applied learning tool that will take the classroom concepts and translate them into practical financial use.

**4. Lessening in Financial Fraud.**

Clear and verified information will minimize fraud and misrepresentation that could affect new investors.

# CHAPTER 3

## MARKET ANALYSIS

### Introduction

An in-depth market study is the key to finding out whether a new business venture is feasible, attractive, and profitable or not, particularly, when the industry is fast-evolving like financial technology (fintech). FinForecaster.pk Pakistan is intending to enter the market that can be described as the increasing rates of digital adoption, low financial literacy and the rising need of easy to access investment tools. The strategic decision-making process requires an awareness of the larger environment, customer behavior, industry trend, and market gap.

The financial-investment landscape in Pakistan is experiencing a massive digital transformation. There has been an ever-increasing pace in internet penetration, smart phone use, online banking, and digital payments in the past ten years. This, in its turn, has provided a favorable field to digital finance solutions and fintech innovations. Although these developments are there, the level of investment in formal capital markets is still low because of the impediments such as complexity, ignorance, and the unavailability of easy-to-use analytical services.

This chapter gives a closer study of the present market environment in terms of market size, investor behavior, gaps in the industry, technology, regulatory factors and competitive environment. It also points out the possibility of positioning FinForecaster.pk Pakistan as a differentiated and value-based solution to investors and learners in the country.

### Financial Market Environment in Pakistan

**Pakistan's Economic Context**

Pakistan is a developing economy and has a population exceeding 240 million and this makes it one of the big consumer markets in the region. Although the economy has been experiencing some problems in terms of inflation, currency depreciation and macroeconomic instabilities, the long-term prospects of the digital financial services are promising due to:

- A young population
- Increasing use of smartphones.

- Raising the consciousness of financial independence.
- Freelancing and entrepreneurship.
- Increasing online banking services.

It is a promising land of fintech products that enhance accessibility and ease the financial decision-making process.

### The Pakistan Stock Exchange (PSX) Environment

The central capital market in the country is the Pakistan Stock Exchange that is instrumental in underwriting investment and financing activities. Regardless of its importance, the participation in PSX is very low. There are only around 300,000-400,000 active investors in a 240 million less than 0.2% of the population making a nation. Conversely, the participation of peers in the region is much higher:

- India: ~4.4% population invested
- Bangladesh: ~3.5%
- Malaysia: ~25%

Such low penetration is not caused by deficiency of market opportunity but their knowledge barriers, unavailability of analytical tools and exposure.

### Increasing Digital Finances Acceptance

In the last five years, Pakistan has witnessed an explosion in the use of the digital:

- More than 90 million broadband subscribers.
- Close to 110 million cell phone subscribers.
- More and more online accounts opened with brokers.

These trends indicate that consumers are becoming more at-home with digital solutions, which preconditions the appearance of such fintech platforms as FinForecaster.pk Pakistan.

## Market Size Estimation

Market sizing can be used to determine the magnitude of the opportunity that FinForecaster.pk Pakistan will be dealing with.

### Total Addressable Market (TAM)

The TAM consists of all the members that may be interested in the platform:

- Retail Investors
- University Students
- Beginner Traders
- Self-employed Professionals and Freelancers.

- Corporate Analysts
- Finance Enthusiasts

According to the number of people who are digitized in Pakistan:

- Internet users: ~70 million
- Smartphone users: ~90 million
- Adults who are interested in finance:
- principles 20-25 million.
- Possible use of fintech: around 10-15 million.

Therefore, TAM of FinForecaster.pk Pakistan is bearing a conservative number of 10-15 million.

## Serviceable Available Market (SAM)

SAM is the realistically reachable population which consists of behavior and interest through digital means:

- Active searchers in the stock market.
- Business news the consumers.
- Commerce and finance programs students.
- People who use bank or investment applications.
- Potential users (estimated): 3-5 million.

## Serviceable Obtainable Market (SOM)

SOM reflects the potential users that can be considered in 2-3 years realistically:

- There are also online brokers in Pakistan with a total of around 300,000-400,000 investors.
- Students in finance/business at universities: between 250,000 and 300,000.
- Freelancers interested in diversifying their investments: up to 100,000+
- Retail investors seeking tools of analysis: 200,000 +

A realistic SOM is 400-600,000 users, and the premium users are at least 30,000+ premium users.

It shows that the platform has a great market potential.

## Market Trends and Customer Behavior.

The way in which Pakistani investors act is important in designing a platform. The modern marketplace can be characterized by a number of trends:

**The Asset Bubble and the Retail Investor Class.**

The new entrants are joining PSX because of:

- Social media influence

- Greater awareness of finances.
- Desire for passive income
- Expansion in the entrepreneurial and freelance business.

Nevertheless, this new category is not knowledgeable or equipped and FinForecaster.pk is very relevant.

**Relevance of Real-Time Information.**

Modern investors prefer:

- Immediate news updates
- Instant price alerts
- Sentiment analysis in real-time.
- The predictions will be made on daily and weekly bases.

The conventional media does not provide information at the right time and digestible fashion.

### Data-Driven Decision-Making

Investors all over the world are increasingly dependent on:

- Machine learning forecasts
- Indicators and signals
- Sentiment scoring
- Predictive analytics

Pakistan is also following some of these trends and this creates an opportunity gap of local AI-based tools.

### Change Fundamental to Hybrid Analysis

There is more and more combination of investors:

- Fundamental metrics
- Sentiment signals
- Forecasting models

FinForecaster.pk is the platform that takes all these tools in one platform, which is not available in Pakistan.

## Industry Voids and opportunities.

**1. Absence of Localized Analysis Power**

Majority of the world forums such as Trading View, Yahoo Finance, and Meta Stock do not offer:

- Localized AI models
- Urdu/English sentiment analysis.
- PSX-specific forecasting

This generates a large gap in the market.

**2. Lack of Introductory Platforms.**

Existing tools are designed to suit well experienced traders, rather than:

- Students
- New investors
- Part-time traders

Simplicity and analytical depth are the main priorities of FinForecaster.pk.

3. **Nonexistent PSX Integrated Ecosystem**.

The investor should verify several locations where:

- Stock prices
- Financial statements
- News
- Predictions
- Ratios

All these factors are combined on a single platform in FinForecaster.pk.

**4. Educational content is Scarce**

No platform in Pakistan provides concomitantly:

- Real-time data
- AI insights
- Beginner lessons
- Examples using real PSX data

There are also universities which do not have applied learning tools. This platform fills that gap.

## Market Attractiveness

The appeal of the market is what defines business idea sustainability.

**1. Growth Potential**

The Pakistani fintech sector can be described as one of the most rapidly expanding sectors. Some of the positive signs are:

- Emerging digital banks in the future.
- Growing investment culture
- Digital adoption by the young generation.
- Increasing online payments

These trends fit perfectly with FinForecaster.pk.

**2. High User Pain Point**

Shareholders are irritated because of:

- Confusing data
- Lack of understanding
- Rumor-driven decisions
- No predictive tools

A company that is dealing with a high pain point has high probability of succeeding.

**3. Low Competition**

Very few platforms offer:

- AI-based PSX forecasting
- News sentiment analysis
- Information +analytics hybrid model.

Therefore, the threat of competitors is low and the opportunity is high.

**4. Regulatory Environment**

The fintech platforms should work keeping in mind the applicable regulations:

**5. Securities and exchange commission Pakistan (SECP)**

SECP regulates

- Brokerage services
- Market data usage
- Corporate governance

FinForecaster.pk is a non-brokerage firm, and thus, the burden of compliance is minimized.

## PSX Regulations

The platform utilizes freely available or licensed data, and therefore compliance is easy.

**1. Data Privacy Framework**

To ensure safe user sign-up and use, there is a need to comply with data protection principles.

On the whole, regulatory risks are moderate to low.

## SWOT Analysis (Market Perspective)

Strengths

- Unique local AI models
- Unified financial and sentiment information.
- Beginner-friendly design
- Predictions and insights in real time.

Weaknesses

- Emergent player in an emerging market.
- Training of algorithms is required.
- Relied on third-party data feeds.

Opportunities

- Huge unexploited online investor pool.
- University partnerships
- Corporate analytics tools
- Regional expansion

Threats

- Data licensing costs
- Foreign fintech platforms entry.
- Disruption of the internet or governmental regulations.

**Conclusions**

Financial intelligence platforms have a very attractive and underserved market in Pakistan. The urgency in this matter is that more and more retailers are involved in PSX, and it is necessary to provide accessible, AI-developed, and friendly analytical tools. FinForecaster.pk Pakistan has a significant market opportunity in terms of their ability to bridge a huge gap in the market by providing a platform which has integrated real-time information, sentiment analysis, forecasting algorithms and educational content that is relevant to Pakistani customer needs.

The gaps in the market and the behavior pattern and market size of the target populations show that the platform can attract a large number of users. Based on a high correlation to digitalization trends, low competition, and high user pain points, the environment is very favorable to the growth of FinForecaster.pk and its sustainability in the long term.

# Chapter 4

## Development and Current Progress

### Requirement Gathering Approach

The systematic and structured manner with which the requirement gathering procedure of this project was undertaken was aimed at guaranteeing the alignment of user expectations, the technical constraints and the academic goals. Instead of using a single method in the requirements elicitation, several requirement elicitation approaches were utilized to provide a holistic view of the problem domain.

Comparative system analysis was one of the major methods employed. The available stock market systems like AKD Trade and JS Invest were analyzed to get a clear insight into their fundamental operations, system processes, and user interfaces. These services are vital in that they offer live market, order placement and simple stock information. But, in this analysis, a number of limitations were realized. Most significantly, such platforms are more oriented towards execution and presentation of raw data, with little depth of analytic information. They lack comprehensive financial analysis, sentiment analysis, and explainable decision support specifically designed to appeal to retail customers and students.

Besides comparative analysis, literature review was also done regarding the financial analytics systems and AI-investment tools. Scholarly literature was also used to determine the best practices in financial data processing, visualization tools, and artificial intelligence to analyze the market. Other trends in financial decision-making emerging in this review include explainable analytics, hybrid analysis of numerical data, and textual data, and the increasing role of sentiment analysis in financial decision-making. Such insights were important in influencing the requirements of the system and making the solution proposed to be future-oriented and relevant.

The other significant component of the requirement gathering process was problem identification which was user-oriented. The requirements and the difficulties of the target users were taken into consideration, which are the retail investors, buyers of the financial process, and the amateur traders. The users of such type are usually not highly financial savvy and find it hard to analyze uncooked information like financial ratios or price fluctuations. They need to be explained, have some visual summaries, and educated instead of using complicated trading tools. The targeting of these groups of users will make sure that the system requirements effectively respond to real-life issues rather than having technical ambition as their sole motivating factor.

In addition, the academic restrictions and project scope were also considered. As it is a Final Year Project and there is a staged development plan, requirements were formulated in a manner that would assist in implementation today and in the future that will give the opportunity to add features like machine learning. Such a moderated balancing has made it possible to stay within the academic schedule with its scalability.

All in all, the requirement gathering methodology which will be used in the given project is a mix of comparative analysis, scholarly research and user-centric design. This approach is important in making sure that the system requirements are clear, realistic, and in line with the educational and practical goals.

## Functional Requirements

Functional requirements specify what the system is supposed to do. They outline the basic features and functionality that the system should offer to meet the needs of users and also accomplish project requirements. Functional requirements in this project are destined to sustain financial analysis, data visualization, transparency and future expansion.

1. **Access Control and Authentication of the User**

The system should be in a position to ensure controlled access by the use of secure user authentication. The platform is supposed to allow users to access the platform with secure credentials. The use of authentication will protect user data and make sure that the personalized features can be supported in the further improvement of the system.

2. **Search and Selection of the Company**

The system should enable customers to query and choose companies traded in the Pakistan Stock Exchange. Users must be provided with an opportunity to find companies with ease, by name or symbol, and gain access to appropriate financial and analytical information. We cannot afford not to have this functionality as we can easily navigate the site.

3. **Company Profile Display**

The system should show in-depth company profile of each chosen company. This entails fundamental details like classification of sector and business description. The company profiles allow the users to know the context of the financial and market data.

4. **Historical Stock Price Trends Visuals**

The system should be able to display past stock price information in chart and graphs. These visualizations provide users with the opportunity to see the price movement and trend developments over time instead of using individual number values. Trend visualization assists in enhancing comprehension of the market trend.

5. **Ratios and Financial Statements Presentation**

The system should display the major financial statements like income statements, balance sheets and the cash flow summaries. According to these assertions, the system ought to compute and show key financial ratios to profitability, liquidity and leverage. This feature aids the fundamental analysis and assists the user to evaluate the performance of the company.

6. **Business news and displaying of Business News**

The system should incorporate business news regarding PSX-listed companies, which will mainly be provided by Business Recorder. The selected company should have relevant news articles that are related to it to give the contextual information about the company as well as numerical data.

**7. Sentiment Visualization**

The system should visualize sentiment indicators based on financial news. These indicators can be positive, negative or neutral sentiment trends. Sentiment visualization supplements financial analysis because it represents market perception and qualitative market factors affecting stock performance.

**8. Dashboard-Based Presentation**

All the outputs of the analytics should be shown in a single dashboard. The financial metrics, price trends, and sentiment indicators should all be integrated on the dashboard. This unified perspective minimizes information overload and increases user knowledge.

**9. Data Management Administrative**

The system should have administrative capabilities of data source, update and system maintenance. This is done so that there is continued data up-to-date and smooth running of the system.

**10. Future Buy, Hold, or Sell Recommendation Support**

Although full implementation of the recommendation modules will be undertaken in subsequent phases, the system has to have the capability of future integration of Buy, Hold or Sell decision logic. This is necessary to make sure that the current design is extensible and ready to use in the future.

## Non-Functional Requirements

Non-functional requirements specify the way the system is supposed to perform. These specifications are aimed at quality features, performance limitation, and operational features which are essential to system success.

**1. Performance**

The system must be responsive to queries and data request of users. Data visualizations, dashboards and charts must be loaded within reasonable response times. User engagement and usability are only possible with good performance.

**2. Scalability**

The architecture of a system must be able to handle the future increase in the volume of data and user traffic. The system should also be able to sustain a consistent performance as the number

of companies, past data, and users increases. Scalability is also especially significant in the future of fintech.

**3. Security**

Security is an essential solution, particularly dealing with user data and financial data. The system should ensure that sensitive data are not accessed by unauthorized individuals and that there is a secure communication between the system parts.

**4. Usability**

The site must be user-friendly and simple to navigate, even by non-technical people. Clearly displayed navigation, clear visualizations, and plain layouts are needed to assist beginner investors and students. Direct impact Usability influences adoption and user satisfaction.

**5. Reliability**

The system must offer the system uniform and reliable functioning, and low downtime. Trustworthiness of systems makes sure that analytical insights are available at the right time.

**6. Maintainability**

The system must be structured in such a manner that it is easy to maintain, debug and add features to it. Long term sustainability and future development are supported as there is clean code structure, modular design and documentation.

## SYSTEM DESIGN

**1. General System Architecture**

System architecture is based on a modular, layered structure to be scalable, flexible and easy to maintain. The architecture is made out of four main layers:

**2. Presentation Layer**

Accountable to user interface, web-based. It presents dashboards, charts and analytical summaries.

**3. Application Layer**

Handles business logic, data processing and frontend to backend service communication.

**4. Data Layer**

Deals with relational and NoSQL database management of structured and unstructured data.

**5. Analytics Layer**

Facilitates analytical processing and also serves as a baseline to future integration of machine learning.

Such a separation of concerns guarantees that changes in one layer will not have negative impacts on the other system parts.

## Workflow Design

The workflow in the system takes a systematic pipeline:

Externals, such as stock and news data, are obtained

- The data collected gets stored in the database based on the schema.
- The data is processed to clean the data and normalize it by using data preprocessing.
- The cleaned data is process by analytical modules.
- Findings are forwarded to the front end to be visualized.
- Dashboards are used to obtain insights

Such workflow can guarantee the efficient handling of data and further future analytical improvements.
**Appendix A has an image of workflow.**

## Database Design

The database architecture is vital towards supporting analytics and scalability. The hybrid approach to database is taken:

1. **Relational Database (PostgreSQL)**

Stored to contain structured information like company information, stock quotations and financial records.

2. **NoSQL Database (MongoDB)**

Maintain poorly structured information like news articles and sentiment related metadata.

The schema is made normal to minimize redundancy and ensure data integrity. Primary and foreign keys are used to define relationships between such entities as companies, financial records, and price data. The model facilitates further storage of machine learning results and score of recommendations.

## User Interface Design

The simplicity and clarity are the main objectives of the user interface. Dashboards provide important measures in the form of charts, tables and other visual aids. It is also easy to navigate, and the user can navigate the companies, financial, and sentiment of the news with less effort. The interface is designed with readability and visual hierarchy in order to improve comprehension in beginner users.

## 1. Methodology and Implementation.

The chapter provides an overview of the approach taken to the design of the proposed analytics platform and the process of how the various elements of the system were incorporated. The methodology is concerned with the data acquisition, storage, preprocessing, and preparation to advanced analytics. Since the project is phased, it is quite clear in this chapter that there are parts that have already been implemented in the current semester, and those that will be implemented in the future. Such a method guarantees transparency in academia and technical rigor.

The approach is tailored to facilitate the overall purpose of the project of offering clear, transparent, and data-driven insights on investments in Pakistan Stock Exchange (PSX) by combining the basic financial analysis and sentiment analysis of the financial news.

## 2. Data Collection

One of the fundamental blocks of the planned system is data collection because the quality and reliability of the results of the analysis directly rely on the quality of the input data. Data collection, in this project, was specifically planned in the light of inconsistent and fragmented information that is usually experienced on various trading websites and applications.

In order to achieve reliability and consistency, two first-hand and authoritative sources of data are used as the system sources: the Pakistan Stock Exchange (PSX) and Business Recorder. The sources chosen were on the basis of credibility, relatability to the financial market of Pakistan, and suitability to the analytical goals of the project.

## 3. Pakistan Stock Exchange (PSX)

Pakistan Stock Exchange is the official provider of stock market data of Pakistan. It entails necessary quantitative data needed in the primary analysis and assessment of historical trends. Information that is gathered on PSX consists of:

- Historical stock prices
- Listings and symbols of companies.
- Sector classifications

Financial reports like income statements, balance sheets and cash flow summaries.

This information is an organized numerical information, which is important to analyses the performance of a company, its financial stability, and long-term growth perspectives. The system makes use of PSX as a major data source which reduces discrepancies that occur when merging information of various trading platforms.

## 4. Business Recorder

Business Recorder is a Pakistani-based financial news that is one of the oldest. It presents publicized and suitable economic and company news in a timely fashion regarding PSX-traded companies. The information available in Business Recorder entails:

- Specific news of the company.
- Economic news on the market.
- Company press releases and news about performance.

This information is a form of unstructured textual information and a significant part of the market sentiment and qualitative variables that drive the behaviors of investors.

**5. Data Characteristics**

The information gathered is both structured and unstructured. Organized information comprises of numerical data like prices and financial ratios whereas unstructured data comprises of textual news items. The methodologies of data collection were to be relevant, consistent and in line with analytical purposes. Information which is not pertaining directly to PSX-listed firms is not gathered in order to keep the focus and analytical acuity.

**Data Storage**

The collected data is stored against a predetermined well-structured database schema. Effective retrieval and analysis of the system, as well as its expansion in the future, require proper data storage.

**Structured Data Storage**

The relational database tables store structured financial data acquired in PSX. This includes:

**Company information**

- Records of the financial statements
- Computed financial ratios

In relational storage, normalization, data integrity and efficient querying all become possible. The relational tables make sure that the company to company, price to price and financial measure relationships are undertaken in the correct manner.

**1. Unstructured Data Storage**

The news articles that are obtained via Business recorder are documented as document-based records. In both documents, there is an article content, date of publication and a company reference. This document-based storage model offers versatility on managing a massive amount of textual information and makes possible any sentiment analysis task in the future.

**2. Separation of Data Types**

The isolation of the structured and unstructured information guarantees the optimized storage and performance. Strict queries are able to be performed well on numerical data and unstructured text data can be processed alone to retrieve sentiment. This design decision would enable the immediate analytical demands as well as the further development of machine learning.

## Data Preprocessing

Raw data when dotted on external sources is usually associated with inconsistencies, missing values, and noise which may adversely impact analysis. Preprocessing of data is therefore an important task in data quality and readiness to analysis.

**1. Financial Preprocessing of Data.**

Preprocessing financial data is aimed at making it more accurate and consistent. The following are the steps undertaken:

**Missing Values:** Records that are missing or incomplete are detected and dealt with accordingly to prevent analytic bias.

**Standardization:** There is standardization of financial values to standard forms and units.

**Checking of Numerical Ranges**: Checking of values is done to make sure they are within reasonable and expected ranges.

**Consistency Checks:** Accompanying financial information between reporting periods is ensured to be consistent.

These measures would confirm that there is no dubious financial data and it is ready to be analyzed in a basic manner.

**2. Textual Data Preprocessing**

Business Recorder textual data will be preprocessed to be sent to sentiment analysis. The preprocessing proceeds are:

**Text normalization**

- Elimination of unnecessary characters and symbols.
- Elimination of redundant whitespace.
- Text structuring to be processed analytically.

These measures enhance the quality of text information and make it fit in the sentiment extraction and analysis process.

## Significance of Preprocessing

Preprocessing of data is critical in the minimization of noise, inconsistency, and enhancement of the outcomes of the analysis. Current analysis and performance of machine learning models are dependent on clean and well-prepared data, which will be used in the future.

**1. Machine Learning Models (Proposed Implementation)**

One of the elements of the long-term vision of this project is machine learning. The machine learning model implementation will however be implemented in the next semester because of academic scheduling and gradual development. The model selection and planning are done in the present stage depending on the system requirements and literature search.

**2. Long Short-Term Memory (LSTM)**

LSTM models are chosen in stock price forecasting because such models have the capacity of modelling sequential and time-dependent data. The price of stock has time-dependence, and therefore LSTM is an appropriate model to incorporate long-term patterns and trends.

In future application, the LSTM models will be trained using historical stock price data obtained at PSX in order to make future predictions on price movements. Such projections will assist in the analysis of information and help in Buy, Hold, or Sell decisions.

**3. Sentiment Analysis FinBERT**

FinBERT will be chosen as a financial sentiment model because it has a domain-specific language understanding. FinBERT is trained on financial text as opposed to general-purpose sentiment models, so it is applicable in the analysis of business and economic news.

FinBERT will be employed to categorize news articles of Business Recorder as positive, negative, or neutral sentiments in the planned implementation. Such sentiment data will be used as qualitative context to the numerical analysis.

**4. Buy, Hold or Sell Prediction Strategy**

The results of the basic analysis and the sentiment analysis will be combined into a decision-support system where Buy, Hold or Sell recommendations will be generated. These recommendations will not only be founded on price forecasts, but they will be factored based on a mixture of:

- Strength indicators financial.
- Market sentiment trends
- Trends in the performance of the past.

This combined methodology makes the recommendations explainable, transparent, and not as much susceptible to the short-term market noise.

## Integration and Evaluation Planning.

The strategies on integration are planned to sustain the process of the seamless integration of machine learning models into the existing system architecture. To measure the performance of models, evaluation metrics like accuracy, trend matching, and correspondence with past results are to be undertaken. Such preparations are necessary, so that the further deployment will become effective and responsible.

## Technology Stack

Choosing a suitable technology stack is such an important part of system development since it directly influences both performance and scalability, maintainability and the user experience. In this project, the technology choice was made with reference to the appropriateness to web-based analytics systems, adherence to the needs of the data processing, and the possibility of integration with machine learning in the future. The selected stack conforms to the project requirements of transparency, scalability, as well as explainable analytics and is viable in an academic development setting.

In this chapter, the frontend, backend, database, and analytical tools to be used in the system are discussed in detail with a planned strategy of deployment.

## Frontend Technologies

The frontend of the system is built on the current JavaScript-based web frameworks like React.js that is popular in the creation of an interactive and reactive web application. These frameworks will allow creating dynamic user interfaces that update effectively as the user interacts with the interface as well as data changes.

React.Js offers a component-oriented system, which enables the interface to be broken down into reusable and maintenance components. The code maintainability is enhanced by this approach, and the complex dashboards can be developed without redundant duplications. Elements (charts, tables, navigation, etc.) can be created separately and smoothly incorporated into the rest of the interface.

These frontend technologies are used to make the platform as responsive and user-friendly as possible regardless of the device used and screen size. This is given that the target audience of the project includes retail investors and students who might be able to access the platform using different devices. Its frontend is designed with simplicity, clarity, and ease of use in mind to cater to novice users.

## Backend Technologies

Python based frameworks like Flask or Fast API are used to implement its backend. These structures prove to be suitable to create lightweight, scalable and effective backend services.

Flask offers a loose and minimalistic architecture that can be used to develop APIs and backend code quickly. It promotes the use of modules and therefore, the various components of the system, including retrieval of data, preprocessing, and analytics, can be easily handled. The simplicity of Flask allows it to be of particular use in academic projects yet remain production-level.

Fast API is taken as an option because of its high performance and the fact that it supports asynchronous activity. It allows to process data faster and effectively handle concurrent requests and this is advantageous to analytics systems that handle multiple user requests at the same time. Fast API has also automatic API documentation which makes it more maintainable and developer productive.

Both of these frameworks are compatible with databases, data processing libraries and machine learning tools. They are used to make sure that the backend is scalable, maintainable, and prepared to be further improved, such as introduction of machine learning models.

## Database Technologies

The system uses hybrid approach of database management to effectively handle various types of data. There are two database technologies, which are **PostgreSQL** and **MongoDB**.

The structured financial information that is stored using PostgreSQL includes the information of companies, historical stock prices, and financial statements, as well as computed ratios. Being a relational database management system, PostgreSQL has high data integrity, complex query support, and high-quality transaction processing. These characteristics are necessary in processing the financial data which must be accurate and consistent.

MongoDB is employed in storage of unstructured data especially the articles of financial news gathered in Business Recorder. News information is not always of the same length and form hence a document-based database is more appropriate. MongoDB can be used to store textual data in a flexible way and facilitate real-time retrieval to be used in sentiment analysis.

Such division between structured and unstructured data enhances the performance and scalability of the systems. It is also of assistance in the future analysis, since data types can be handled with the methods that are most appropriate to the structure of the data type.

## Deployment Strategy (Planned)

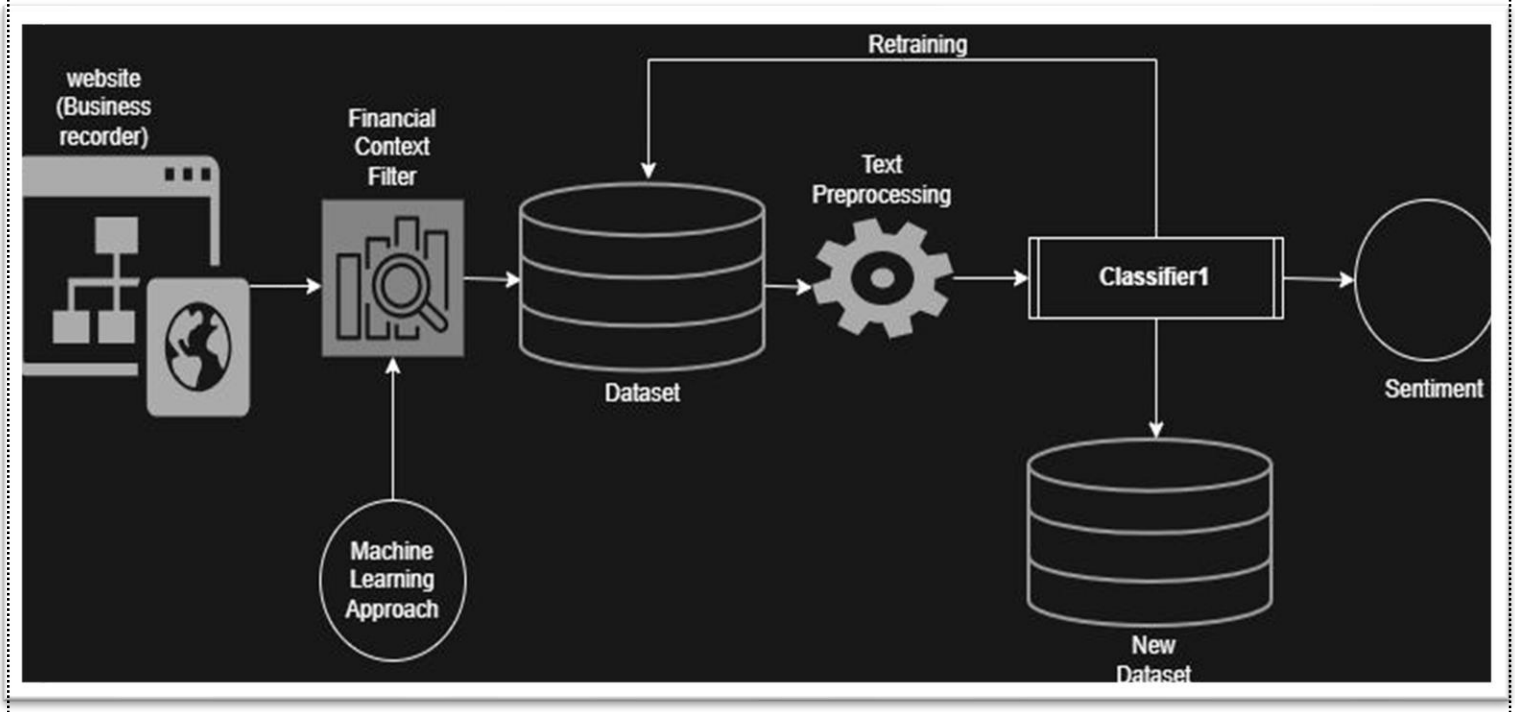
It is also built keeping the concept of the cloud-based deployment strategy to enhance scalability, availability, and performance.

Cloud deployment has a few benefits among them such as on-demand resource allocation, high availability, and easy scaling as the demand by the users increases. It also facilitates centralized information management and dependability in monitoring of systems. The present system architecture is also to be cloud compatible despite the fact that full deployment will be completed at a later stage.

The project plans and sets up cloud deployment early enough, which means that the platform is capable of scaling to a real-life fintech application in the future once starting with an academic prototype.

# Appendix

## Appendix A:

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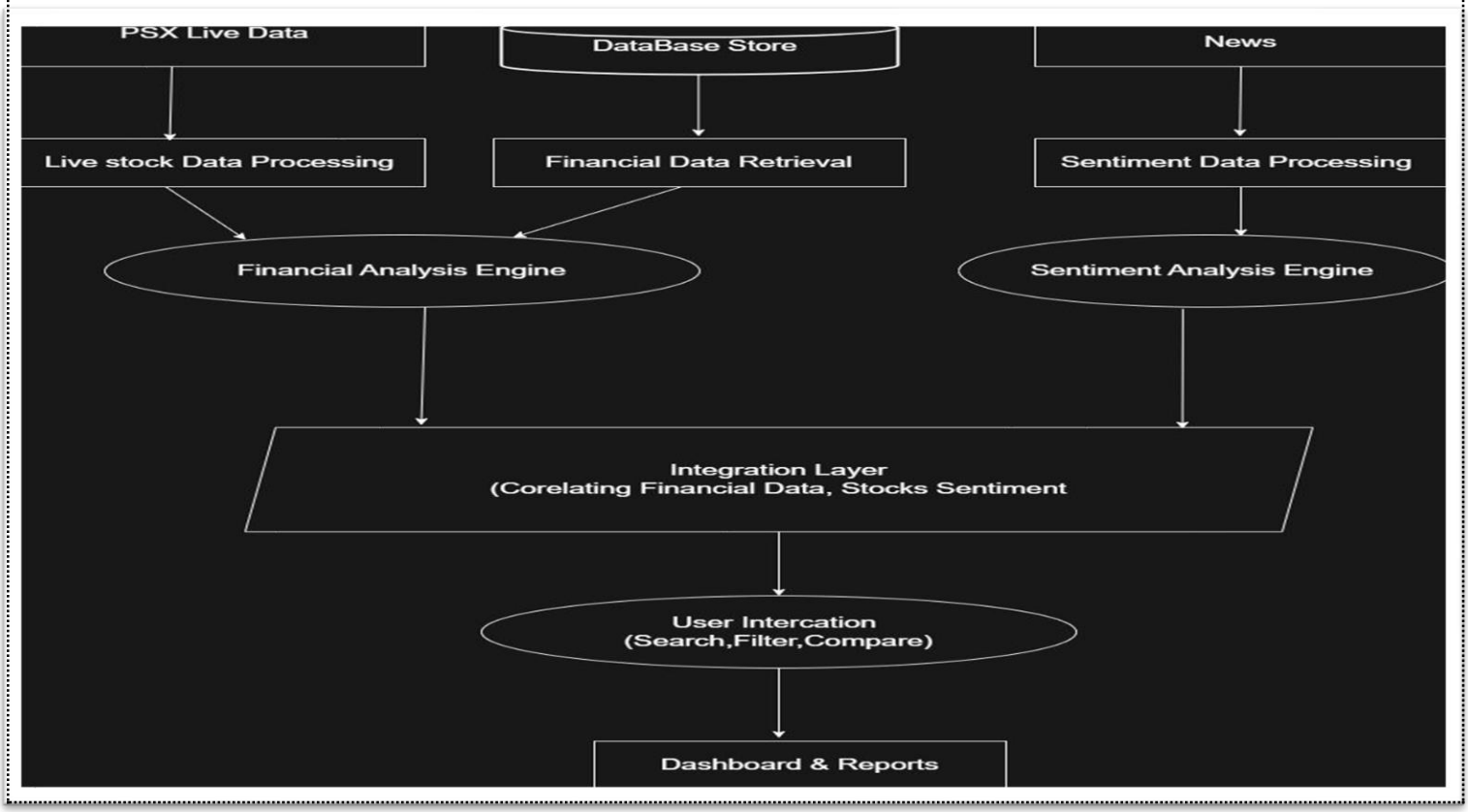

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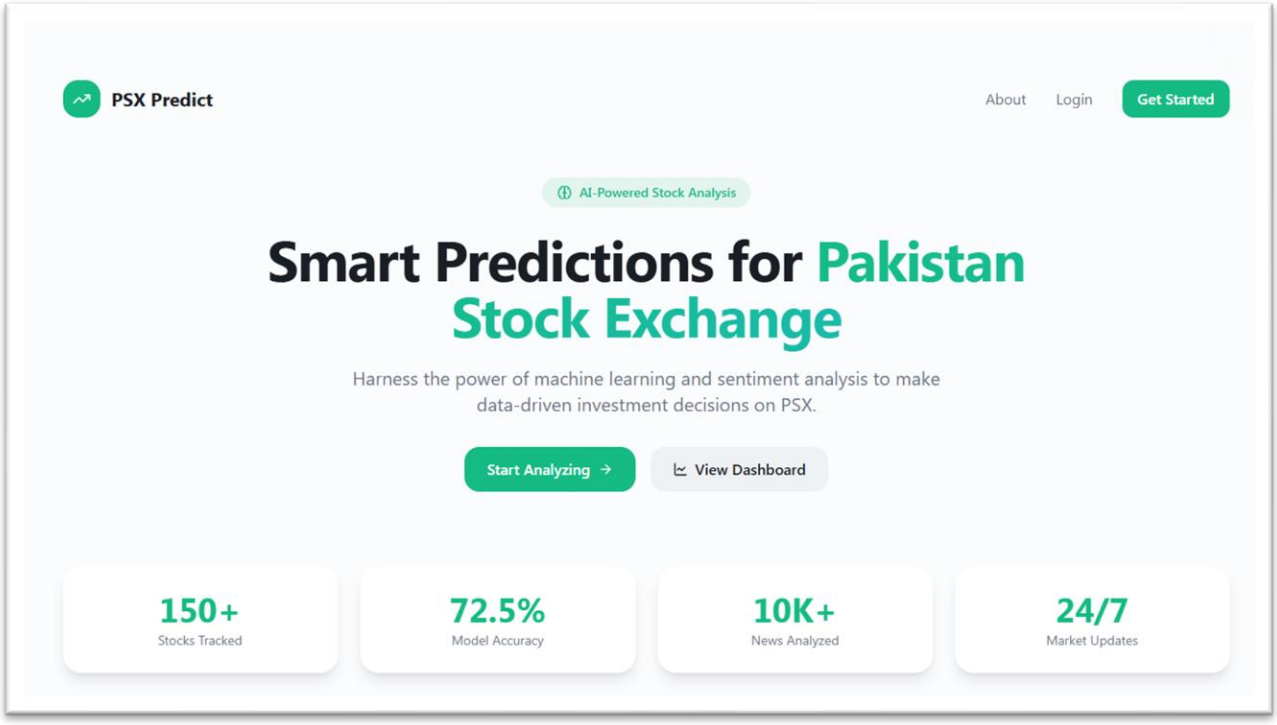
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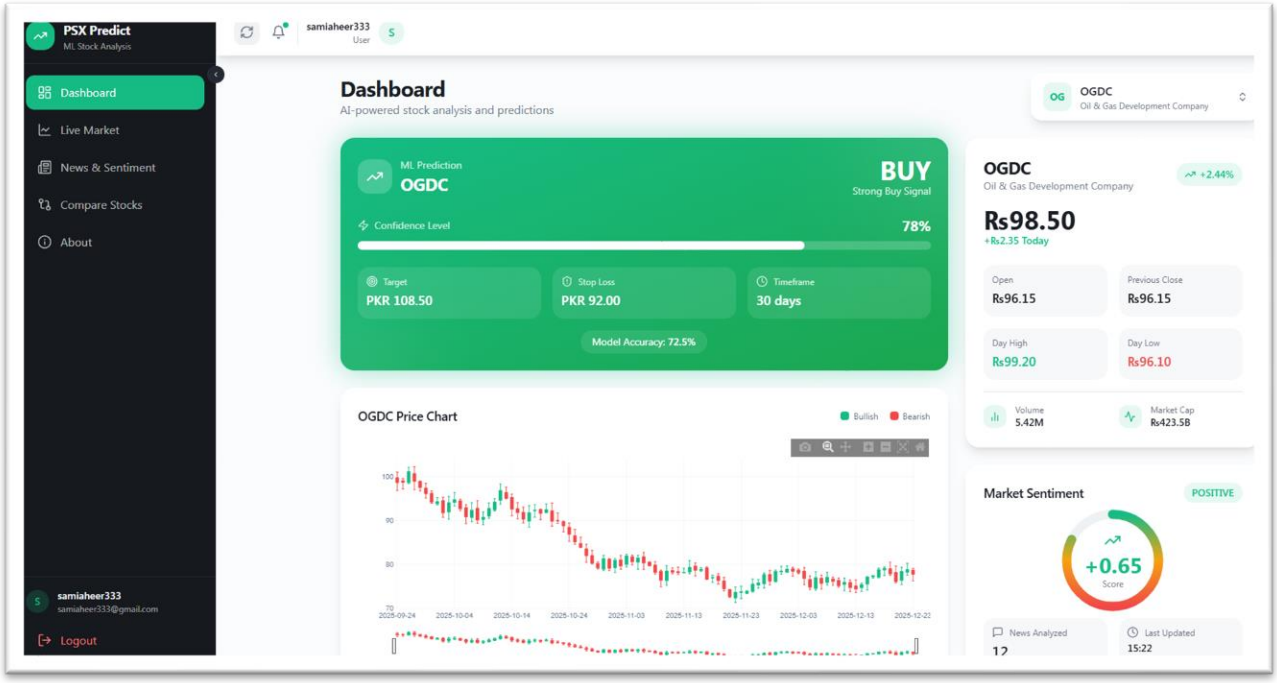
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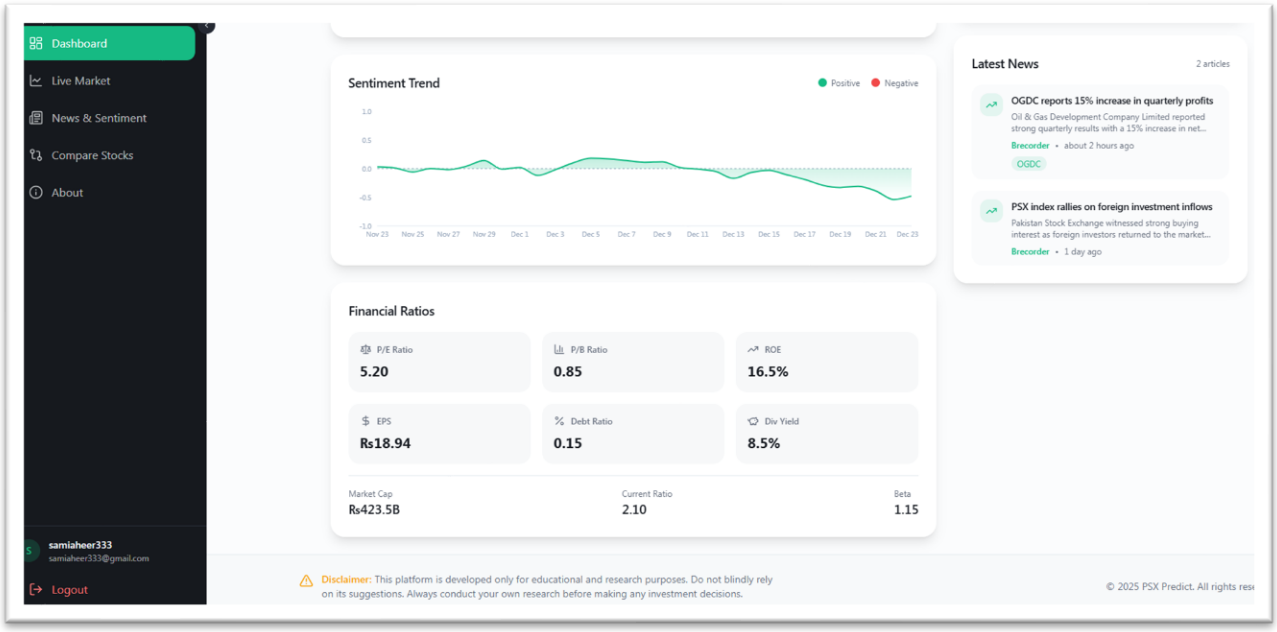
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2 articles
OGDC reports 15% increase in quarterly profits
Oil & Gas Development Company Limited reported strong quarterly results with a 15% increase in net...
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PSX index rallies on foreign investment inflows
Pakistan Stock Exchange witnessed strong buying interest as foreign investors returned to the market...
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Financial Ratios
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P/B Ratio
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EPS
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Debt Ratio
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Div Yield
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Market Cap
Rs423.5B
Current Ratio
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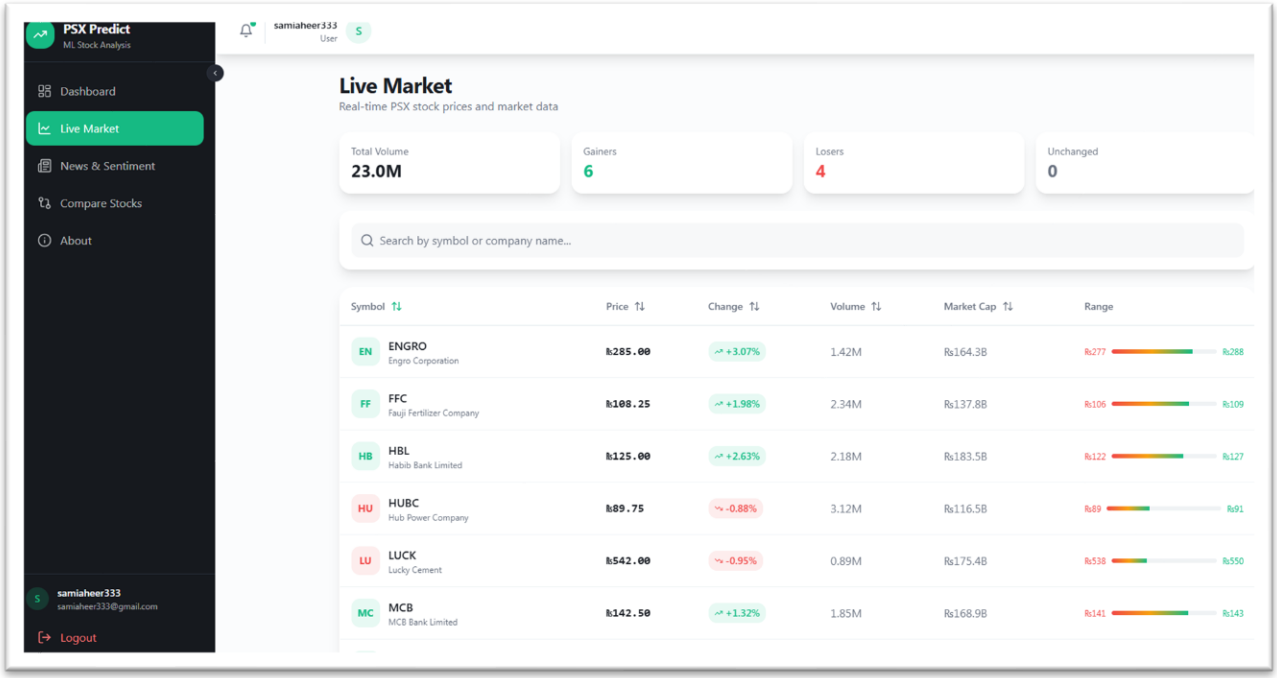
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Gainers
6
Losers
4
Unchanged
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Symbol
Price
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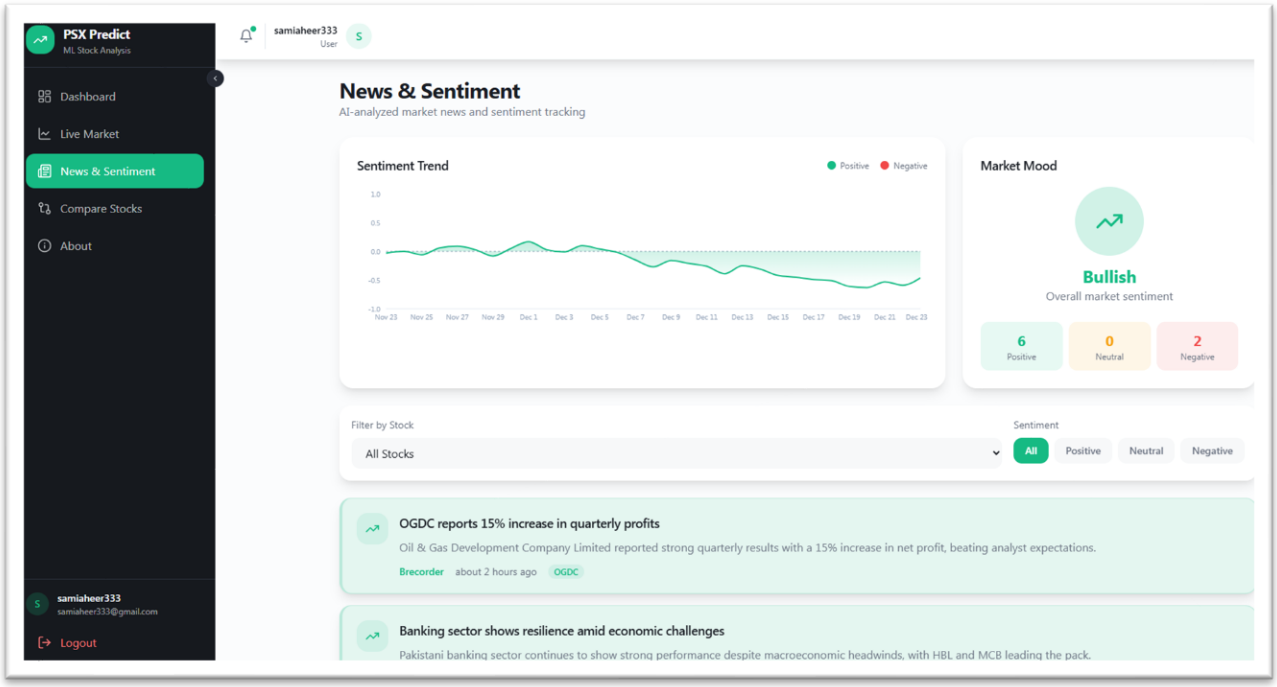
Volume
Market Cap
Range
EN ENGRO Engro Corporation ₨285.00 +3.07% 1.42M Rs164.3B Rs277 Rs288
FF FFC Fauji Fertilizer Company ₨108.25 +1.98% 2.34M Rs137.8B Rs106 Rs109
HB HBL Habib Bank Limited ₨125.00 +2.63% 2.18M Rs183.5B Rs122 Rs127
HU HUBC Hub Power Company ₨89.75 -0.88% 3.12M Rs116.5B Rs89 Rs91
LU LUCK Lucky Cement ₨542.00 -0.95% 0.89M Rs175.4B Rs538 Rs550
MC MCB MCB Bank Limited ₨142.50 +1.32% 1.85M Rs168.9B Rs141 Rs143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Neutral
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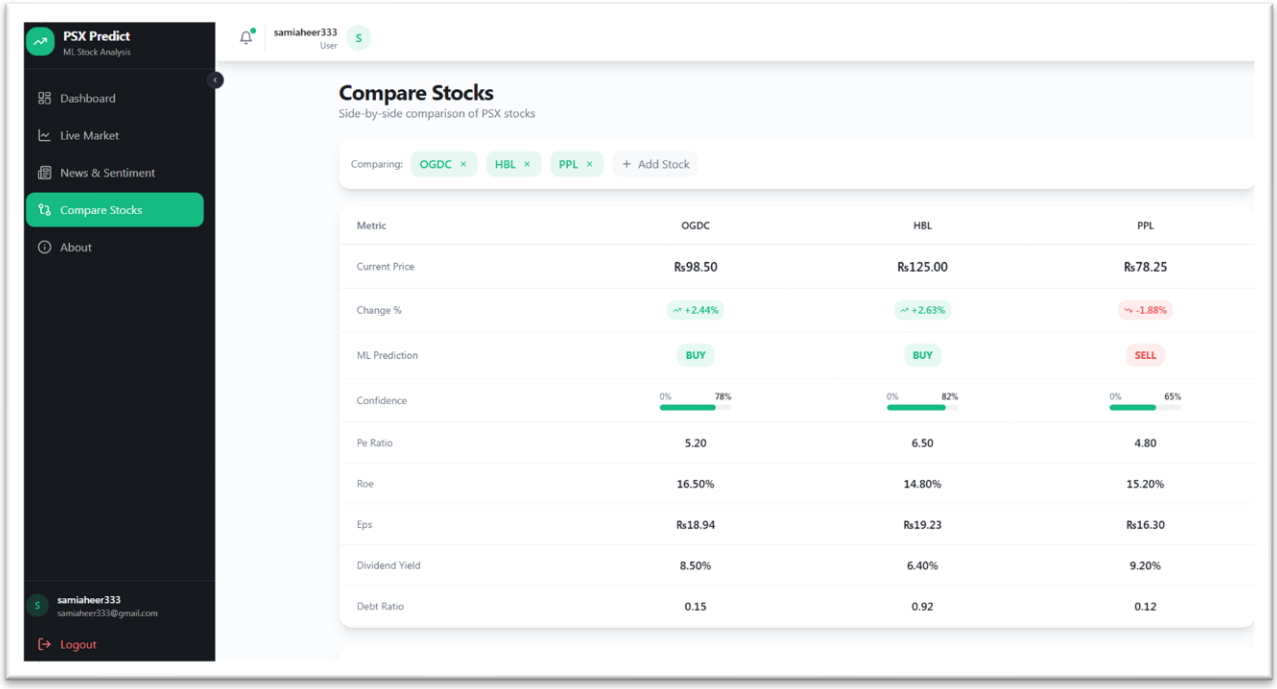
Banking sector shows resilience amid economic challenges
Pakistani banking sector continues to show strong performance despite macroeconomic headwinds, with HBL and MCB leading the pack.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Compare Stocks
Side-by-side comparison of PSX stocks
Comparing: OGDC × HBL × PPL × + Add Stock
Metric | OGDC | HBL | PPL
Current Price | Rs98.50 | Rs125.00 | Rs78.25
Change % | +2.44% | +2.63% | -1.88%
ML Prediction | BUY | BUY | SELL
Confidence | 0% 78% | 0% 82% | 0% 65%
Pe Ratio | 5.20 | 6.50 | 4.80
Roe | 16.50% | 14.80% | 15.20%
Eps | Rs18.94 | Rs19.23 | Rs16.30
Dividend Yield | 8.50% | 6.40% | 9.20%
Debt Ratio | 0.15 | 0.92 | 0.12